Math 111: Calculus Homework 4

Due Friday, November 13 by 10:00P.M. via Gradescope

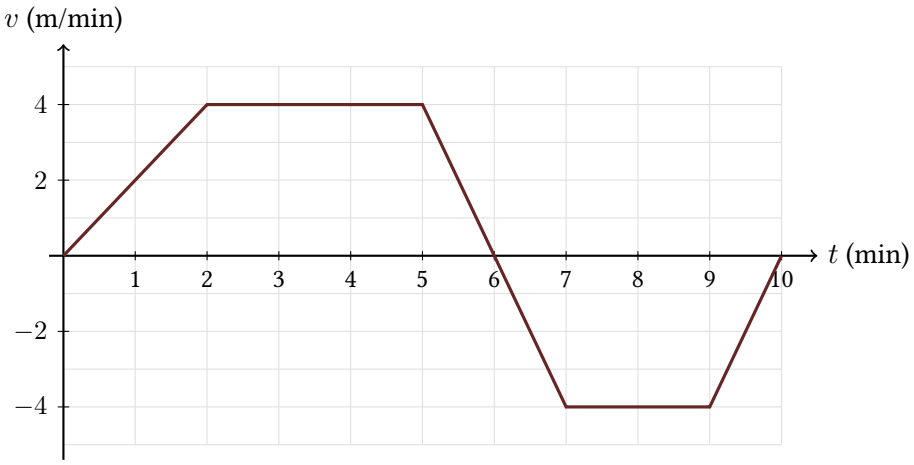
Covers Active Calculus §4.1, §4.2, §4.3, and §5.6. Problems 1 and 3 use §4.1–§4.3 and can be done after Wednesday, November 4. Problem 2(a)–(b) uses §4.2; Problem 2(c)–(e) and Problem 4 use §5.6 (Friday, November 6). No problem needs the Fundamental Theorem of Calculus; if you know it, you may use it to *check* an answer, but explain each answer the way the problem asks.

Before you start. Write in sentences and paragraphs, as an explanation another student in this course could read and follow. Draw diagrams large and label them. You are encouraged to work with classmates; cite everyone you worked with, and write your solutions independently — duplicated solutions are an Honor Principle violation. Each problem is marked holistically on the 0–5 scale, with up to two further points for the quality of the writing. If you need an extension, use the extension form *before* the deadline.

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1. A drone's altitude, from its velocity

A survey drone takes off from a platform and flies straight up and down for ten minutes. Its vertical velocity $v(t)$, in meters per minute, is graphed below; positive means rising. The graph is made of straight segments.



Let $h(t)$ be the drone's height above the platform, in meters, at time t, so $h(0) = 0$.

(a) On which time intervals is the drone rising, and on which is it falling? How do you know from the graph?

(b) How high is the drone at $t = 6$? Explain why that is the greatest height it reaches.

(c) Where is the drone at $t = 10$, relative to the platform?

(d) What total distance does the drone travel over the ten minutes? Explain in a sentence why this is not the same as your answer to (c).

(e) Find the drone's average velocity over $[0, 10]$, with units. Then draw, on a copy of the graph, a horizontal line at that height, and say what the picture shows about the area above and below it.

(f) Sketch a graph of $h(t)$ on $[0, 10]$. Label the height at $t = 2, 5, 6, 7, 9, 10$. Say where h is concave up, where it is concave down, and where it is a straight line, and say how you read each of those off the graph of v.

2. Filling a tank from a table

Water flows into a tank at a rate $r(t)$, in liters per minute, measured every five minutes:

t (min)	0	5	10	15	20
$r(t)$ (L/min)	2	3	5	8	12

Assume throughout that r is increasing on $[0, 20]$. The quantity we want is the amount of water that flows in over the twenty minutes, $\displaystyle\int_0^{20} r(t)\,dt$.

(a) Compute the left and right sums L_4 and R_4, with units. Which is an over-estimate and which is an under-estimate? Explain using a picture, not a rule.

(b) Explain why, for an increasing r and n equal subintervals, $R_n - L_n = \dfrac{20}{n}\big(r(20) - r(0)\big)$. Then suppose the rate could be measured as often as you like. How many equal subintervals would guarantee that the average $\frac{1}{2}(L_n + R_n)$ is within 1 liter of the true amount? How often is that, in seconds?

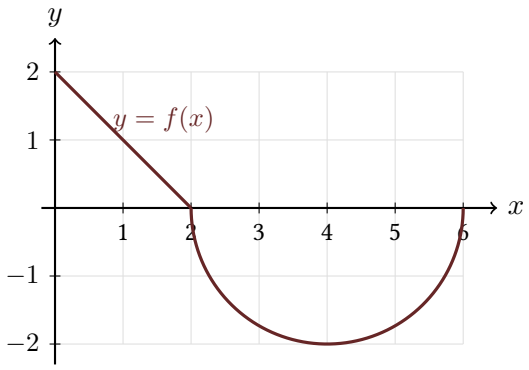
(c) Compute the trapezoid estimate T_4. The table suggests r is concave up (look at the differences). Assuming it is, is T_4 too big or too small? Explain by drawing the graph of r over a single subinterval together with the trapezoid on that subinterval.

(d) Use the readings at $t = 5$ and $t = 15$ to compute the midpoint estimate M_2 (two subintervals of width 10). Assuming r is concave up, is M_2 too big or too small? Explain by drawing the graph of r over a single subinterval together with the midpoint rectangle on that subinterval. What interval does your work in (c) and (d) trap the true amount in?

(e) Compute T_2, and then Simpson's estimate $S = \dfrac{2M_2 + T_2}{3}$. Does it land in your interval from (d)? In a sentence or two, explain why combining M and T with these weights should be better than either one alone.

3. Integrals you can read off a picture

The function f on $[0, 6]$ is graphed below. On $[0, 2]$ it is a straight line; on $[2, 6]$ it is the lower half of a circle of radius 2 centered at $(4, 0)$.



Use geometry and the properties of the definite integral; no Riemann sums are needed.

(a) Find $\displaystyle\int_0^2 f(x)\,dx$, $\displaystyle\int_2^6 f(x)\,dx$, and $\displaystyle\int_0^6 f(x)\,dx$. Explain why one of them is negative.

(b) Find $\displaystyle\int_0^6 |f(x)|\,dx$ and $\displaystyle\int_6^0 f(x)\,dx$.

(c) Find $\displaystyle\int_0^6 \big(3f(x) + 1\big)\,dx$. Say which properties of the integral you used.

(d) Find the average value of f on $[0, 6]$. Is it positive or negative? Explain in a sentence why that matches the picture.

(e) Suppose $f(x)$ is the rate, in thousands of dollars per month, at which a small business's bank balance changes, x months after opening. Interpret your answers to (a) in a few sentences, with units: what happened over the first two months, over the next four, and over all six?

4. When is an approximation exact?

(a) A student submits the following.

> **Claim.** The trapezoid rule always gives the exact value of an integral.
>
> **Argument.** Left sums always under-estimate an integral, and right sums always over-estimate it. The trapezoid estimate T_n is the average of L_n and R_n. The average of an under-estimate and an over-estimate is exact. So T_n is always equal to the integral.

Identify the *first* sentence of the argument that is wrong, and say what is wrong with it. Then show that the third sentence is wrong too, even for a function where the first sentence is true: give a specific increasing function, interval and n where T_n is not exact, and explain how you know without computing the integral.

(b) Give a function for which T_n *is* exact for every n, and explain why.

(c) Let $f(x) = x(4 - x)$ on $[0, 4]$. Compute L_4 and R_4. Explain why $L_4 = R_4$ here, by comparing which values of f each sum uses. Then explain why L_4 is nevertheless *not* equal to $\int_0^4 f(x)\,dx$, and say whether it is too big or too small.

(d) By drawing a line over a single subinterval together with the midpoint rectangle on that subinterval, explain why the midpoint rule M_n gives the exact value of $\int_a^b (mx + c)\,dx$ for every linear function and every n.